King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If the vectors $\vec{u} = \langle 1, a, b\rangle$, $\vec{v} = \langle a, b, 1\rangle$, and $\vec{w} = \langle b, 1, a\rangle$ lie in the same plane, then

(a) $a^2 + b^2 = 3ab + 1$

(b) $a = b$

(c) $a^3 + b^3 = 3ab - 1$

(d) $a^3 + b^3 = ab - 2$

(e) $a^3 - b^3 = 2ab$

2. Which one of the following statements is TRUE about the graph given by the parametric equations

$$x = t^2 - t, \quad y = 2t^3 - 6t + 5$$

(a) It has one horizontal tangent and two vertical tangents

(b) It has one horizontal tangent and no vertical tangent

(c) It has no horizontal tangent and two vertical tangents

(d) It has one horizontal tangent and one vertical tangent

(e) It has two horizontal tangents and one vertical tangent

3. Which set of the following parametric equations represents a parabola or part of a parabola?

(a) $x = 1 + \cos t,\ y = \sin t,\ t \in [0, 2\pi]$

(b) $x = t + 1,\ y = \sqrt{t},\ t \geq 0$

(c) $x = \sec t,\ y = \cos t,\ t \in \left[0, \dfrac{\pi}{2}\right)$

(d) $x = \dfrac{1}{t},\ y = t - 1,\ t \in (0, \infty)$

(e) $x = 2t - 1,\ y = 2t + 1,\ t \in (-\infty, \infty)$

4. The curve represented by the parametric equations

$$x = e^{-t}, \quad y = e^{3t}, \quad t \geq 0$$

is represented by the rectangular equation

(a) $y = x^3,\ 0 < x \leq 1$

(b) $y = \dfrac{1}{3x},\ x \geq 1$

(c) $y = \dfrac{1}{x^3},\ x \geq 1$

(d) $y = \dfrac{1}{\sqrt[3]{x}},\ x \geq 1$

(e) $y = \dfrac{1}{x^3},\ 0 < x \leq 1$

5. The vector with initial point $P(1, 1, -3)$ and terminal point $Q(3, 4, -4)$ is parallel to the vector

(a) $\langle 2, -9, 3 \rangle$

(b) $\left\langle 1, \dfrac{3}{2}, -1 \right\rangle$

(c) $\langle 1, 1, -3 \rangle$

(d) $\left\langle -\dfrac{2}{3}, -1, \dfrac{1}{3} \right\rangle$

(e) $\langle 3, 4, -4 \rangle$

6. The area of the parallelogram that has the vectors $\vec{u} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{v} = \vec{j} + \vec{k}$ as adjacent sides is equal to

(a) $\dfrac{\sqrt{2}}{2}$

(b) $3\sqrt{2}$

(c) 2

(d) $\sqrt{2}$

(e) $2\sqrt{2}$

7. If a point has polar coordinates $(r, \theta) = \left(-4, -\dfrac{3\pi}{4}\right)$, then the corresponding rectangular coordinates for the point are

(a) $(x, y) = (2\sqrt{2},\ 2\sqrt{2})$

(b) $(x, y) = (-2\sqrt{2},\ 2\sqrt{2})$

(c) $(x, y) = (-2\sqrt{2},\ -2\sqrt{2})$

(d) $(x, y) = (2\sqrt{2},\ -2\sqrt{2})$

(e) $(x, y) = (\sqrt{2},\ \sqrt{2})$

8. The area of the region inside the polar curve $r = 4\cos\theta$ and outside the polar curve $r = 2$ is equal to

(a) $\dfrac{2\pi - \sqrt{3}}{6}$

(b) $\dfrac{3\pi + \sqrt{3}}{6}$

(c) $\dfrac{4\pi + 6\sqrt{3}}{3}$

(d) $\dfrac{4\pi + 5\sqrt{3}}{3}$

(e) $\pi + \dfrac{\sqrt{3}}{2}$

9. The projection of $\vec{u} = \langle 0, 3, 3\rangle$ onto $\vec{v} = \langle -1, 1, 1\rangle$ is

(a) $\langle -3, 3, 3\rangle$

(b) $\langle -1, 4, 4\rangle$

(c) $\langle -2, 2, 2\rangle$

(d) $\langle 0, 1, 4\rangle$

(e) $\langle 2, -2, 2\rangle$

10. The area of the region inside the graph of the polar equation $r = 2\sin\theta$ for $\dfrac{\pi}{4} \leq \theta \leq \dfrac{\pi}{2}$ is equal to

(a) $\dfrac{\pi}{4} + \dfrac{1}{2}$

(b) $\dfrac{\pi}{4} + 1$

(c) $\dfrac{\pi}{2} + \dfrac{1}{2}$

(d) $\dfrac{3\pi}{4} - \dfrac{1}{2}$

(e) $\dfrac{\pi}{2} - \dfrac{1}{2}$

11. If $R(a, b, c)$ is the point that lies two-thirds of the way from $P(1, 0, 1)$ to $Q(0, 2, 0)$, then $a + b + c =$

(a) 2

(b) $\dfrac{4}{3}$

(c) 4

(d) $-\dfrac{2}{3}$

(e) -3

12. An equation of the tangent line to the parametric curve

$$x = t^2 - 4, \quad y = t^2 - 2t$$

at the point $(x, y) = (-3, 3)$ is given by

(a) $y = 3$

(b) $y = x + 6$

(c) $y = 2x + 9$

(d) $x = -3$

(e) $y = -2x - 3$

13. The surface area of the surface generated by revolving the parametric curve

$$x = \cos^2 t, \quad y = 2\cos t, \quad 0 \le t \le \frac{\pi}{2}$$

about the x-axis is given by

(a) $\int_0^{\pi/2} 8\pi \sin t \cdot \sqrt{1 + \cos^2 t}\, dt$

(b) $\int_0^{\pi/2} 4\pi \sin(2t) \cdot \sqrt{1 + \cos^2 t}\, dt$

(c) $\int_0^{\pi/2} 2\pi \cos^2 t \cdot \sqrt{1 + \cos^2 t}\, dt$

(d) $\int_0^{\pi/2} 2\pi \cos(2t) \cdot \sqrt{1 + \cos^2 t}\, dt$

(e) $\int_0^{\pi/2} 8\pi \cos t \cdot \sqrt{1 + \cos^2 t}\, dt$

14. The length of the polar curve

$$r = 2 - 2\cos\theta, \quad 0 \le \theta \le \frac{\pi}{3}$$

is equal to

(a) $6 - \sqrt{3}$

(b) $8 - 4\sqrt{3}$

(c) $4 + \sqrt{3}$

(d) $4 + 2\sqrt{3}$

(e) $2 - 2\sqrt{3}$

15. The graph of the polar equation $r = \dfrac{6}{\sin\theta - 3\cos\theta}$ is

(a) a line

(b) a hyperbola

(c) a parabola

(d) a circle

(e) an ellipse

16. The center C and the radius r of the sphere

$$x^2 + y^2 + z^2 - 2x + 6y + 8z + 1 = 0$$

are

(a) $C(-1, 3, 4),\ r = 5$

(b) $C(1, 3, -4),\ r = 3$

(c) $C(1, -3, -4),\ r = 5$

(d) $C(-1, -3, 4),\ r = 5$

(e) $C(-2, 6, 8),\ r = 4$

17. The slope of the tangent line to the polar graph $r = 3 - 3\cos\theta$ at the point corresponding to $\theta = \dfrac{\pi}{4}$ is

(a) $\dfrac{1 - \sqrt{2}}{\sqrt{2}}$

(b) -1

(c) $-\dfrac{1}{\sqrt{2}}$

(d) $\dfrac{\sqrt{2}}{2 - \sqrt{2}}$

(e) $\dfrac{-\sqrt{2}}{2 + \sqrt{2}}$

18. Which one of the following vectors is perpendicular to the tangent line to the graph of $f(x) = x^4 - 2x$ at the point $(1, -1)$?

(a) $\langle 2, -1\rangle$

(b) $\langle -2, 4\rangle$

(c) $\langle 1, 2\rangle$

(d) $\langle 2, 1\rangle$

(e) $\langle -1, 1\rangle$

19. The curve given by the parametric equations

$$x = 1 + t^2, \quad y = 1 + t^2 + t^3$$

is concave upward when

(a) $t < 0$

(b) $-\infty < t < \infty$

(c) $-1 < t < 1$

(d) $t > -2$

(e) $t > 0$

20. Question 10 of the original exam is omitted (missing from the published exam file).

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\det = 3ab - 1 - a^3 - b^3 = 0$, so $a^3 + b^3 = 3ab - 1$.
2	**(e)**	$dy/dt = 0$ at $t = \pm 1$ (2 horizontal); $dx/dt = 0$ at $t = \frac{1}{2}$ (1 vertical).
3	**(b)**	$t = y^2$ gives $x = y^2 + 1$, part of a parabola; others are line, circle, hyperbolas.
4	**(e)**	$y = x^{-3}$ and $x = e^{-t} \in (0, 1]$ for $t \geq 0$.
5	**(d)**	$\overrightarrow{PQ} = \langle 2, 3, -1\rangle = -3\langle -\frac{2}{3}, -1, \frac{1}{3}\rangle$.
6	**(d)**	$\vec{u} \times \vec{v} = \langle 0, -1, 1\rangle$, area $= \|\vec{u} \times \vec{v}\| = \sqrt{2}$.
7	**(a)**	$x = -4\cos(-\frac{3\pi}{4}) = 2\sqrt{2}$, $y = -4\sin(-\frac{3\pi}{4}) = 2\sqrt{2}$.
8	**(c)**	Limits $\pm\frac{\pi}{3}$: $A = \int_0^{\pi/3}(16\cos^2\theta - 4)d\theta = \frac{4\pi+6\sqrt{3}}{3}$.
9	**(c)**	$\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\vec{v} = \frac{6}{3}\langle -1, 1, 1\rangle = \langle -2, 2, 2\rangle$.
10	**(a)**	$A = \int_{\pi/4}^{\pi/2}(1 - \cos 2\theta)d\theta = \frac{\pi}{4} + \frac{1}{2}$.
11	**(a)**	$R = P + \frac{2}{3}\overrightarrow{PQ} = (\frac{1}{3}, \frac{4}{3}, \frac{1}{3})$, so $a + b + c = 2$.
12	**(c)**	$t = -1$, slope $\frac{2t-2}{2t} = 2$: $y - 3 = 2(x + 3)$, so $y = 2x + 9$.
13	**(b)**	$S = \int 2\pi y\, ds$ with $ds = 2\sin t\sqrt{1 + \cos^2 t}\, dt$ gives $4\pi \sin 2t\sqrt{1 + \cos^2 t}$.
14	**(b)**	$r^2 + r'^2 = 16\sin^2\frac{\theta}{2}$, $L = \int_0^{\pi/3} 4\sin\frac{\theta}{2}\, d\theta = 8 - 4\sqrt{3}$.
15	**(a)**	$r\sin\theta - 3r\cos\theta = 6 \Rightarrow y - 3x = 6$, a line.
16	**(c)**	Completing squares: $(x - 1)^2 + (y + 3)^2 + (z + 4)^2 = 25$.
17	**(d)**	$\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta} = \frac{\sqrt{2}}{2-\sqrt{2}}$.
18	**(a)**	Tangent direction $\langle 1, 2\rangle$; $\langle 2, -1\rangle \cdot \langle 1, 2\rangle = 0$.
19	**(e)**	$\frac{d^2y}{dx^2} = \frac{3}{4t} > 0$ when $t > 0$.
20	omitted	Omitted.